

Exploring the Relationship Between Urban Greenery and Crime Rates in New York City

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ABSTRACT

Urban greenery has always been associated with enhancing quality of life in a city, but its impact on crime and safety remains a topic of debate (Kim, Young-Jae, and Eun Jung Kim). In this study, we explore the relationship between trees and crime across New York City zip codes using data from the 2015 NYC Tree Census and crime records. We have used clustering algorithms to analyze whether tree-related factors like tree health, density, and location could predict higher-than-average crime rates. The results showed healthier trees correlated with lower crime rates and lower crime penalties. These insights can help city planners use greenery to build safer cities. Future research in this field could entail exploring more detailed spatial data and temporal trends to deepen the understanding between the relationship between urban trees and crime rates.

KEYWORDS

Urban greenery, crime rates, New York City, tree health, tree density, clustering algorithms, K-Means, decision tree classifier, heatmaps, spatial analysis, urban planning, tree diameter, crime patterns, urban trees, safety, community well-being, machine learning, PCA, feature engineering, green spaces.

1. INTRODUCTION

The idea of a green city is an exciting concept – a bustling metropolis that promotes health not only for its residents but for the planet as well. Trees in cities produce oxygen and provide shelter from the sun and wind, but how do they impact crime rates? The purpose of this study is to investigate the correlation between trees and crime among New York City zip codes and narrow in on the most impactful factors within that relationship. To do so, we combined a crime dataset and a tree dataset on zip code in NYC from 2015 and studied the ability of clustering algorithms’ abilities to determine if a zip code has higher than average amounts of crime based on the tree data provided. What we found was striking: healthier, well-maintained trees tended to align with lower crime penalties, supporting the idea that urban green spaces can

contribute to safer, more liveable neighbourhoods. This study builds on earlier research performed by others and uses modern machine learning techniques to investigate how urban greenery impacts crime. It also highlights the importance of approaching these findings carefully. Planting trees in an underserved neighbourhood might reduce crime over time, but it could also drive-up housing costs and displace residents who can no longer afford to live there. By exploring these ideas further, we hope to encourage city planners and policymakers to strike a thoughtful balance. Urban greenery can improve environmental quality, enhance community well-being, and—when done right—make cities feel safer and more connected. Future work could build upon these findings by integrating additional datasets, such as socioeconomic indicators, housing density, and demographic patterns, to provide a more comprehensive picture of urban dynamics. Moreover, conducting similar analyses in other cities would help determine whether the observed patterns are unique to New York City or generalizable across different urban environments. Long-term studies that measure crime and community development before and after the implementation of green initiatives would also be valuable in assessing their true impact. The rest of this paper explores the details of our approach, findings, and their implications for creating greener and safer cities. Ultimately, this research underscores the potential of urban greenery to serve as a catalyst for positive change. When integrated thoughtfully into urban planning, trees can improve air quality, enhance public spaces, and contribute to safer, more vibrant communities. By leveraging these insights, cities can take meaningful steps toward creating environments where nature and people coexist harmoniously, fostering a better quality of life for all.

2. RELATED WORK

There were 3 studies in the cities of Philadelphia PA, Baltimore MD, and Portland OR that all concluded that trees generally contribute to reduced rates of crime. Specifically, trees that are tall enough to not obstruct the view from house windows to streets, and trees that seem taken care of reduce crime the most (Conniff, 2012). On the other hand, trees that obstruct views of the street can

increase crime by giving criminals a place to hide. Trees and other healthy greenery give the impression that the neighborhood residents take an active role in their community and care about their surroundings. The study performed in Baltimore found a direct correlation to tree canopy size and a reduction in crime, which has a strong correlation with tree height and health (Troy et al., 2012). Beyond the correlation between crime reduction and tree height and health, trees benefit urban communities by filtering air, improving residents' mental health, and providing shelter from the elements ("Speak for the Trees, Boston", 2024). We aim to build on these findings using New York City data from 2015.

3. METHODOLOGY

3.1 Data Cleaning and Feature Engineering

In this section, we describe the steps we took to clean our initial tree and crime datasets and transform them into one dataset with useful information using zip code as the index. We acquired the New York City tree census data from Kaggle. The original dataset includes data collected in 2015, 2005, and 1995 by NYC Parks and Recreation staff, TreesCount! program staff, and hundreds of volunteers. The NYC crime data was acquired from the NYC Open Data website, which is dedicated to making data produced by the New York City government easily and freely accessible to the public. Because the only year both datasets had in common was 2015, we chose to limit the data to be within 2015.

3.1.1 Data Cleaning

Our first step in making the data ready for machine learning models was data cleaning. First, we extracted the features in each dataset we found relevant to the problem at hand. The data we kept for the tree dataset was tree Diameter at Breast Height (DBH) in meters, curb location (either on curb or off curb), tree health (alive or dead), tree common species name, sidewalk damage measure (damaged or not damaged), and zip code. For the crime dataset, we kept the violation date and time, the crime issuing agency, dollar amount of penalty imposed, the charge codes for up to 10 levied charges, and zip code. Since we are interested in each crime, not each crime report, we used the pandas melt function to turn each non-null charge code column into its own row. After paring down the datasets to just the relevant features, we removed null values. We removed every row with a null value from the tree dataset, going from 750352 rows originally to 595662. Before we ran the melt function on the crime dataset to break it down to one row per crime, there were 750352 rows. After running the melt function there were 7503520 rows, and after removing rows with null values after melting there were 414769 rows.

3.1.2 Feature Engineering

Because we needed our final dataset structure to have one row per zip code, we aggregated data for all rows that related to the same

zip code. To do this, we grouped each dataset by zip code then ran built-in and custom aggregation functions to engineer new columns. Our first custom aggregation strategy calculates a ratio for columns with dichotomous categorical data like the tree dataset sidewalk damage column by dividing the count of one of the two possible values by the count of the other value. Our second custom aggregation strategy breaks a single nominal categorical column down into 3 separate columns, one for the most listed value in the category, one for the second most listed value, and one for the third most listed value. Including the top 3 values in a nominal categorical column as their own columns allows the model to take frequency into account without getting overfit on all the options in the column. Beyond the custom aggregation strategies, we added columns to the tree dataset for count of trees by counting the rows per zip code and mean tree DBH by averaging the tree DBH column per zip code. We added a mean penalty-imposed column and a total crime count column for the crime dataset, both per zip code.

3.1.3 Final Data Cleaning Steps

After both the tree and crime datasets were set up with unique zip codes and new features that aggregated information in a meaningful way within each zip code, we encoded the columns that were produced from the first/second/third prevalence functions in a way that ensured they mapped to the same value in all 3 columns. Those were the only categorical columns so they were the only ones we needed to encode before running models. We chose the SciKitLearn LabelEncoder because our non-numeric data is nominal categorical data with no inherent order and we wanted to maintain the dimensionality of the dataset.

To combine the tree and crime datasets we performed an inner join on zip code because we wanted the final dataset to only include zip codes that both datasets had. Once we scaled the full dataset after the join so that all data was in a comparable number range, the data was ready for the machine learning models.

4. EXPERIMENTS

In this section, we explore how tree characteristics relate to crime rates across the New York City zip codes. These experiments focus on two main goals: 1. understanding the patterns in the data through visualizations and 2. using the machine learning models to evaluate whether tree-related factors can help predict crime trends. We started with visual analysis of input variables to discover existing relationships and then trained clustering and classification models to dig deeper into these relationships. The next steps break down how we set up our experiments, the approaches we took, and what we discovered.

4.1 Models

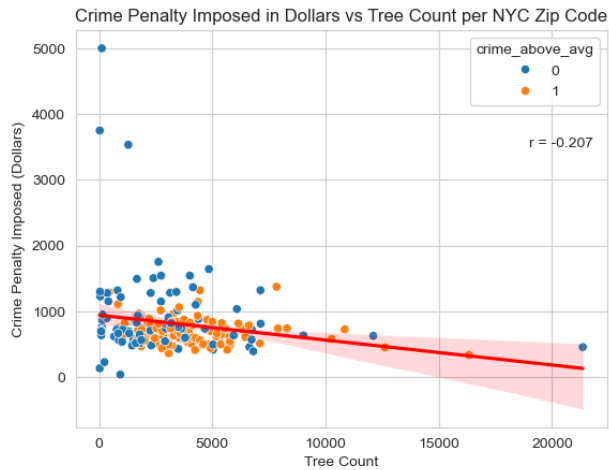


Figure 1: Crime Penalty vs Tree Count per Zip Code

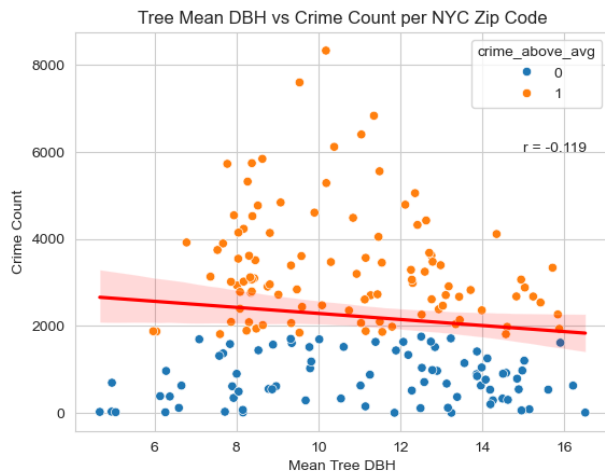


Figure 2: Mean Tree DBH vs Crime Count per Zip Code

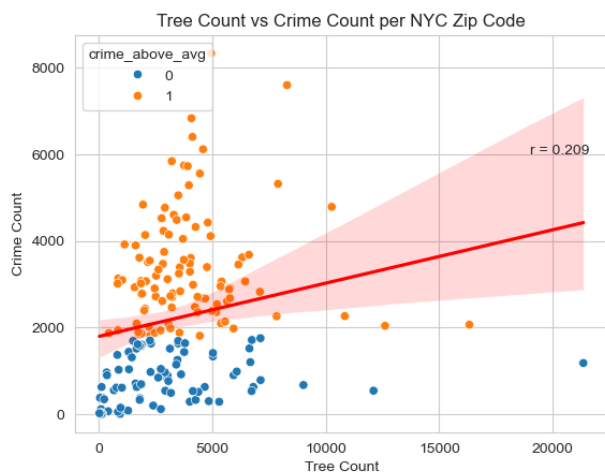


Figure 3: Tree Count vs Crime Count per Zip Code

We plotted and analyzed three scatterplots (Figures 1, 2, and 3) to explore the relationship between the tree characteristics and the crime patterns before running models. The points were color-coded orange if the crime rates were above average and blue if they were below average. Figure 1 examines the correlation between the total number of trees and the penalties imposed for the crimes, with a regression line showing a slight negative trend. This suggests that the areas with more trees tend to have lower crime penalties, possibly reflecting less severe crimes in these locations. Figure 2 examines the average tree diameter at breast height (DBH) in meters versus the total crime count, but the regression line has a correlation coefficient (r) of -0.119 , indicating essentially no amount of linear relationship between two features. Figure 3 analyzes the relationship between the tree count and the total crime count. The line of best fit for this scatterplot has a correlation coefficient of 0.209 , indicating a very low amount of linear relationship between these two variables as well. These three visualizations highlight the obvious linear relationship between crime penalty imposed and tree count and the lack of a clear linear relationship between tree mean DBH, tree count, and crime count per NYC zip code.

4.2 K-Nearest Neighbors

We used the K-Nearest Neighbors (KNN) classification algorithm to see how well tree attributes could predict whether a zip code had above-average crime rates. First, we divided the dataset into features (X) and the target variable (y), where X contained all columns except `crime_above_avg`, and y represented whether crime rates were above or below average. The dataset was then split into training and testing sets, with two-thirds allocated for training and one-third for testing. Next, we trained the KNN model using a range of values for the number of neighbors (k), from 2 to 20, to identify the optimal value for the algorithm.



Figure 4: Finding Optimal K for K Nearest Neighbors Model

For each value of k , we evaluated the model using 5-fold cross-validation, tracking the model performance metrics of F1-Score,

Precision, and Recall. The results were plotted in Figure 4 to visually assess how these metrics changed with increasing values of `k`. Based on the evaluation, the optimal `k` was found to be 4, as it provided the best balance across all metrics, maximizing the weighted F1-Score. Once the optimal `k` was determined to be 4, the model was trained using this value and tested on the held-out testing dataset. The final model evaluation using 5-fold cross-validation yielded an average F1-Score of 0.6487, Precision of 0.6745, and Recall of 0.6603, indicating fairly strong overall performance. To further interpret the model's predictions, a confusion matrix (Figure 5) was generated, which shows the breakdown of correct and incorrect classifications for above-average and below-average crime rates. The confusion matrix indicates that the model was better at predicting higher than average crime rate (24 accurate classifications) than lower than average crime rate (16 accurate classifications), and it gave false positive results (says the crime rate is above average when it should be below) slightly more often than false negative results. The high number of false negatives suggests the model struggles to identify above-average crime rates consistently, while false positives indicate occasional overprediction.

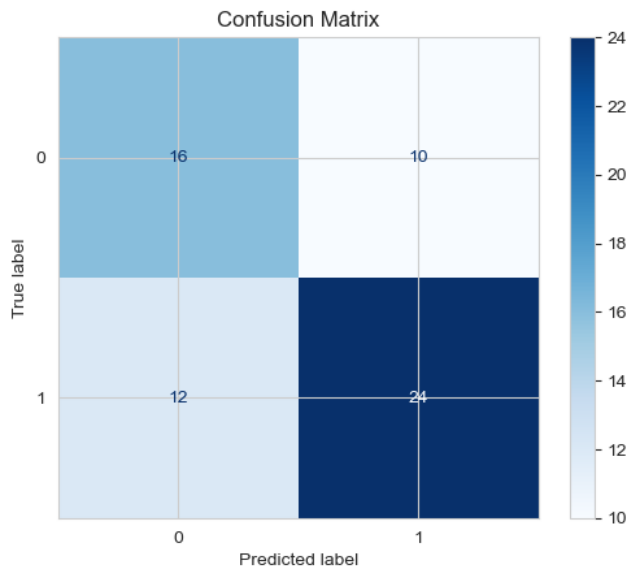


Figure 5: KNN Confusion Matrix

These results demonstrate that the KNN algorithm, with the selected features and optimal parameters, can effectively identify patterns in the data and classify crime levels with reasonable accuracy.

4.3 K-Means Clustering

We implemented the K-Means clustering algorithm to group zip codes based on their tree and crime characteristics. This helped us better understand the patterns in the data.

K-Means clustering is an unsupervised machine learning algorithm used to group data into K distinct clusters based on the features of the dataset. The algorithm iteratively minimizes the distance between points and their respective cluster centers. The goal of K-Means clustering is to minimize the total intra-cluster variance or the sum of squared distances between each point and its corresponding cluster centroid. This is expressed as -

$$J = \sum_{k=1}^K \sum_{i=1}^{n_k} \|X_i - c_k\|^2$$

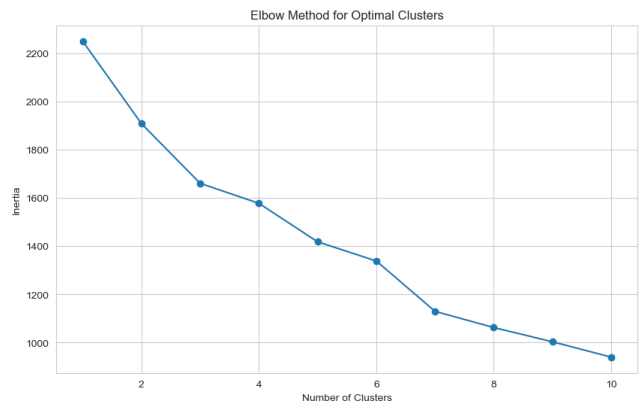


Figure 6: Elbow Method for Optimal Clusters

First, we determined the optimal number of clusters using the Elbow Method (Figure 6). This involved testing cluster numbers ranging from 1 to 10 and calculating how tightly data points were grouped within each cluster (measured by inertia). The results showed that three clusters struck the right balance, as adding more clusters did not significantly improve performance.

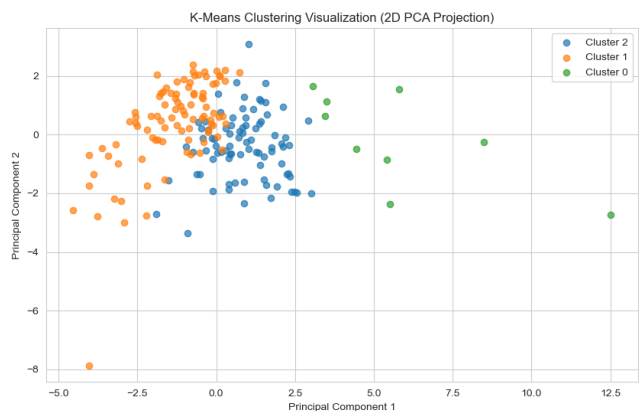


Figure 7: PCA for K-Means Clustering Results

To visualize the clusters more effectively, we used Principal Component Analysis (PCA) to reduce the dataset's complexity and project into two dimensions. The PCA plot (Figure 7) shows how

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well the data points group into three distinct clusters with a bit of overlap between cluster 1 and 2, making it easier to see where the data is distinguishable into clusters and where it may get predictions wrong due to cluster overlap. After identifying the optimal number of clusters, we trained the K-Means model with three clusters and assigned each zip code to one of these groups. These cluster assignments were then used to explore the relationship between tree health and crime rates.

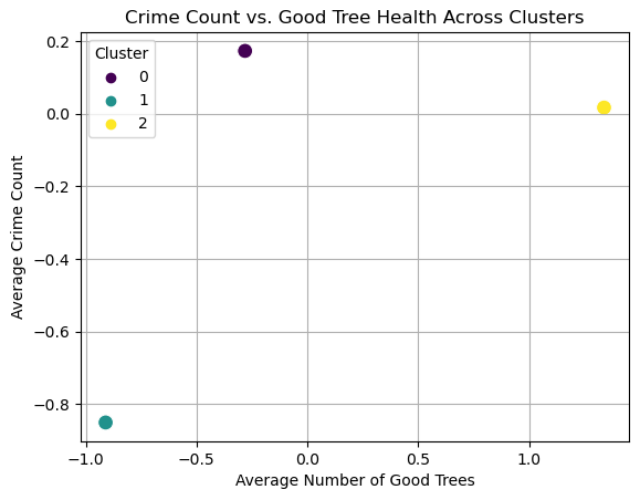


Figure 8a: Avg Number of Good Trees vs Average Crime Count

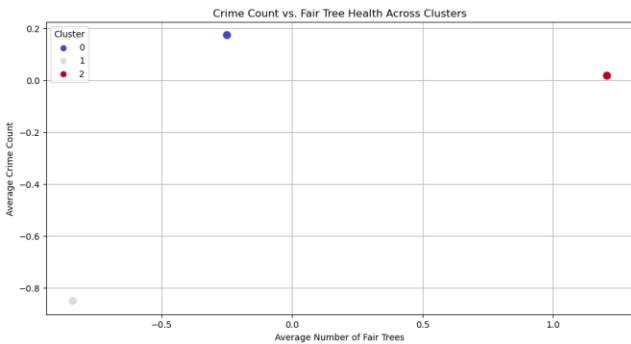


Figure 8b: Avg Number of Fair Trees vs Avg Crime Count

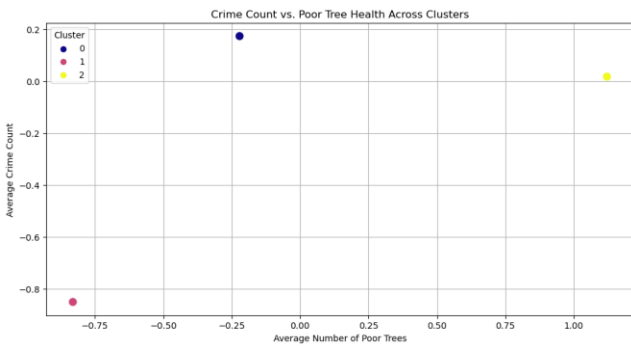


Figure 8c: Avg Number of Poor Trees vs Avg Crime Count

Figure 8a, 8b and 8c show how the average number of good, fair, and poor trees in each cluster related to crime counts, visually depicting how different levels of tree health correlated with crime patterns across clusters.

5. RESULTS

5.1 Heatmaps

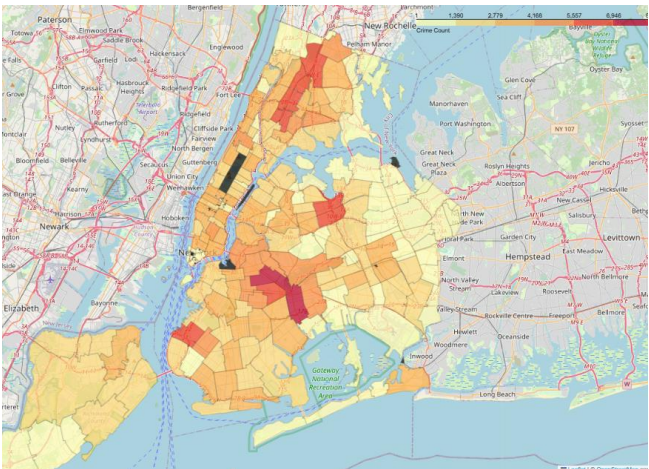


Figure 9: Heatmap 1 - Crime Count Heatmap

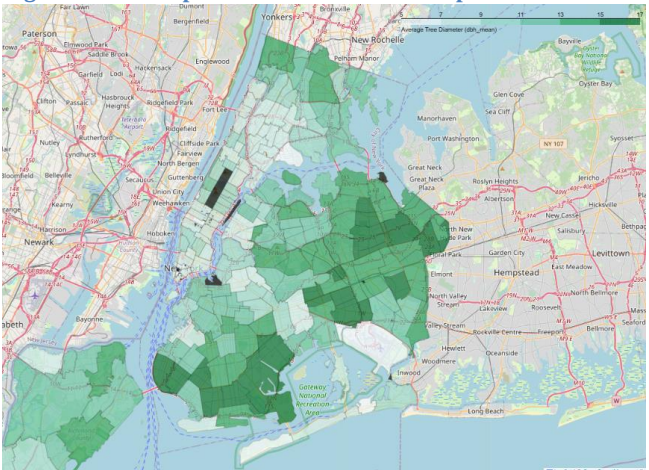


Figure 10: Heatmap 2 – DBH Mean Heatmap

To analyze the spatial distribution of crime and tree characteristics across New York City, we created two heatmaps that combine geographic data with zip-code-level attributes. These visualizations were generated by merging a GeoJSON file containing NYC zip code boundaries with a dataset of attributes, including crime counts and tree-related metrics such as average tree diameter at breast height (dbh_mean). The centroids of the zip code geometries were used to calculate the center of the map, ensuring that the visualizations were properly aligned and focused. The first heatmap displays crime counts across NYC zip codes, using shades of red to highlight areas with higher crime rates. This visualization allows us to identify crime hotspots and their

geographic spread. Areas with the darkest shades of red represent zip codes with the highest reported crime counts, typically concentrated in urban hubs or economically challenged neighborhoods. This information can be used by policymakers and law enforcement to strategically allocate resources and address areas with higher crime prevalence.

The second heatmap focuses on the average tree diameter (dbh_mean) across NYC zip codes, represented in shades of green. Darker green areas indicate zip codes with larger tree diameters, which often correspond to older, healthier, or more well-maintained urban trees. These areas may also reflect neighborhoods with better access to green spaces or higher investment in urban forestry initiatives. Conversely, areas with lighter green shades suggest a lack of mature or healthy trees, potentially correlating with more urbanized or underdeveloped regions.

These heatmaps provide valuable insights into the interplay between urban greenery and crime. Regions with higher average tree diameters, as shown on the tree heatmap, often align with zip codes exhibiting lower crime counts on the crime heatmap. This supports findings from previous analyses suggesting that urban green spaces and healthier trees may contribute to reduced crime rates and enhanced community well-being. By visualizing these relationships, the maps emphasize the potential of strategic urban planning to create safer and more livable environments.

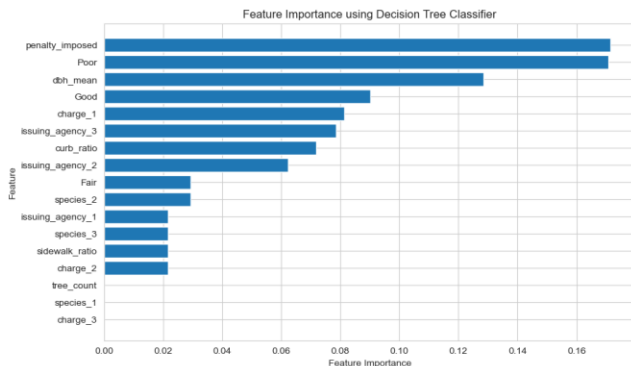


Figure 11: Decision Tree Classifier Feature Importance

We performed a decision tree classifier to examine the relative feature importances of our input features to whether a zip code had above average crime. The 4 most important features to the training of the model, in order, are crime penalty imposed, whether the tree was in poor health, DBH mean, and whether a tree was in good health. Crime penalty imposed is almost as important to the model as the tree being in poor health, implying a similar strength correlation between crime rates and penalty imposed and tree health.

5. DISCUSSIONS

According to the results discussed in the Experiments section, there is a correlation between higher tree diameter at breast height (DBH)

and lower crime rates. This is similar to the findings made in Troy et al. that taller trees with broader canopies are correlated with lower crime rates. Additionally, the findings of the feature importance graph from the decision tree classifier reveals that trees being in good condition play a large role in the prediction of crime. City planners could use these results in designing a safer city. They could choose to plant more mature trees, thus increasing the DBH and height of the tree from the get-go, and perhaps choose trees that take less maintenance to stay in good condition. Renters and homebuyers can also look for large, well-cared-for trees as an indicator of a safer neighborhood. The heatmaps (Figures 9 and 10) can serve as essential tools for urban planners, policymakers, and community stakeholders. The crime heatmap highlights specific zip codes requiring targeted interventions to address crime, while the tree diameter heatmap underscores areas where investments in urban forestry could improve environmental quality and potentially reduce crime. Together, these visualizations offer a data-driven foundation for decision-making, emphasizing the importance of balancing urban development with the preservation and expansion of green spaces to foster healthier and safer communities. While the outcomes of this study are useful, it is important to keep the ethical implications in mind. Less safe neighborhoods tend to be more affordable places to live, so if city planners plant trees in an effort to reduce crime it may drive out residents who can't afford to live elsewhere. The article "Trees Shed Bad Rap As Accessories to Crime" by Richard Conniff includes a quote from USFS researcher Geoffrey Donovan that essentially points out that a burglar alarm will be much more effective than trees as a safety measure. Trees are not a tool to keep people safe and that in an unsafe neighborhood, but they can be used to improve the quality of life for the people living in the neighborhood. To continue the work of this study, we'd like to increase the amount of tree data we use in model training. Adding canopy cover, tree height, lowest branch height, and perhaps even oxygen emission data would uncover even more intricate relationships between urban trees and crime rate. It would also be beneficial to get more granular spatial data and perhaps get the rows down to blocks or street sections as opposed to just zip codes. Finally, this is a study that is best conducted over time, and it would be interesting to study crime rate in an urban area with freshly planted trees.

6. CONCLUSION

This study highlights the interesting connection between urban greenery and crime rates, showing how healthier trees and green spaces are often linked to safer neighborhoods. We have used clustering techniques, heatmaps, and decision tree models and found out that areas with bigger, well-maintained trees tend to have lower crime rates. These findings support the idea that urban green spaces can contribute to creating safer and more livable communities. However, it is important to approach this insight carefully. (Sypion, Natalia). While trees can enhance quality of life, they work best alongside other measures like better policing and community programs. Looking ahead, adding more tree-related

data, like canopy cover or tree height, and analyzing neighborhoods at a more detailed level could help uncover even deeper insights. Long-term studies on how crime rates change after tree-planting initiatives would also be valuable. By balancing these efforts thoughtfully, city planners and policymakers can create greener, safer, and more inclusive cities for everyone.

7. ACKNOWLEDGEMENTS

We are grateful to the New York City Government for providing open access to the datasets that made this research possible. The 2015 NYC Tree Census and OATH ECB Hearings Case Status datasets were essential for understanding the relationship between urban greenery and crime rates across NYC neighborhoods. We also appreciate the efforts of the NYC Open Data platform and Kaggle for maintaining such well-organized and accessible resources that support research and innovation.

We would also like to thank our advisor for the guidance and thoughtful feedback throughout the project. This input played a key role in shaping our analysis and improving the overall quality of this study.

8. AUTHOR CONTRIBUTIONS

Teammate A Erica Schermerhorn)	Teammate B (Yash Kulkarni)
Researched potential topics and explored urban safety challenges, focusing on urban greenery and crime rates.	Researched potential topics and explored urban safety challenges, focusing on urban greenery and crime rates.
Preprocessed the data by handling missing values and developed custom feature engineering algorithms.	Preprocessed the data by handling missing values
Implemented K-Nearest Neighbor clustering algorithm model for crime count and fair, poor and good tree. Health.	Implemented K-Means clustering algorithm model for crime count and fair, poor and good tree. Health.
Created graph visualizations for initial data analysis and post K-Means outcome analysis.	Created interactive choropleth maps to visualize crime count and tree health metrics (good, fair, and poor tree conditions) across NYC zip codes using GeoJSON data and Folium.

9. DATA AND CODE AVAILABILITY

9.1 Datasets

1. Tree Census in New York City - <https://www.kaggle.com/datasets/nycparks/tree-census>
2. OATH ECB Hearings Case Status in New York City - https://data.cityofnewyork.us/City-Government/FY-2015/xpju-hibv/about_data
3. nyc-zip-code-tabulation-areas-polygons GeoJSON file https://github.com/fedhere/PUI2015_EC/blob/master/mam1612_EC/nyc-zip-code-tabulation-areas-polygons.geojson

9.2 Code Availability

The code is accessible using the following link - <https://code.vt.edu/erica16/urban-computing> and we have also submitted on Canvas.

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